Anurag Mahapatra

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Full-stack engineer focused on privacy-preserving cryptographic systems, AI behavioral tooling and multi-agent orchestration. Shipped BlinkBin, a zero-knowledge ephemeral secret-sharing tool with client-side AES-256-GCM encryption and a cryptographically enforced drand IBE time-lock, alongside a 2-model extractor-judge sycophancy detector that intercepts ChatGPT outputs in under 300 ms. Delivered a production PWA for 400+ conference check-ins and maintained cloud infrastructure for 3,000+ students, spanning the full stack from zero-knowledge architecture through containerized deployment, event-driven pipelines and CI/CD.

Technical Skills

Languages	Python, JavaScript, Java, C/C++, SQL, Bash
AI/ML	LangChain, RAG (FAISS), NLP (VADER, TF-IDF), Hugging Face, OpenAI & Anthropic SDKs
Web Technologies	React, Node.js, Express, FastAPI, MongoDB, PostgreSQL, Supabase, WebSockets, REST APIs, JWT
DevOps & Cloud	Docker, AWS (EC2, S3, RDS, Lambda, CloudWatch), Nginx, Redis, Kafka, GitHub Actions, CI/CD, Linux
Tools & Frameworks	Git, Cloudflare, OAuth 2.0, Agile/Scrum

Technical Projects

BlinkBin – Zero-Knowledge Ephemeral Secret Sharing — GitHub | Live

- Architected a zero-knowledge paste platform with AES-256-GCM client-side encryption and URL-fragment-only key delivery; the server is inherently incapable of decryption, and the Dead Drop IBE time-lock via drand Quicknet makes early decryption mathematically impossible before the user-set time, not merely policy-restricted
- Deployed on a single EC2 t3.micro via Docker Compose (FastAPI, Redis AOF+RDB, Kafka, Expiry Worker) behind Cloudflare Pages and Tunnel; 12 configurable paste modes across lock status, burn condition and PBKDF2-wrapped passwords (100k iterations), with a 30-day hard expiry ceiling and Kafka-driven deletion decoupled from Redis TTL

Drift – LLM Sycophancy Detection Chrome Extension — GitHub

- Built a Chrome extension that detects unjustified stance shifts in LLM responses with ~150–300 ms latency, intercepting outputs before final render across ChatGPT
- Developed a 2-model extractor-judge architecture where the Extractor performs stance isolation and new-information detection independently before passing distilled summaries to a Judge model; reduces token usage by ~50–60% versus single-model full-context approaches while keeping detection behavior consistent

DebateItOut – Context-Engineered AI Debate Engine — GitHub

- Engineered an AI orchestration engine that splits available models into 2 opposing factions and drives structured multi-turn debates on a user-defined proposition; uses context engineering to enforce per-faction context isolation, preventing cross-faction contamination across debate turns
- Implemented a script-controlled moderator layer with context-aware routing that compresses per-turn arguments before dispatch; abstraction covers Anthropic and OpenAI-compatible APIs including Ollama with ~200–400 ms model-switch latency, keeping extended debates coherent as context grows

Experience

Freelance Full Stack Developer – International Conference on Smart & Sustainable Urbanism — Mar 2026 – Apr 2026

- Developed a secure meal ticketing system (PWA, Supabase) to process and validate check-ins for ~400+ attendees; automated QR generation workflows and delivered a real-time analytics dashboard

Education & Certifications

B.Tech in Computer Science & Engineering – CGPA: **8.77/10.0** — Expected May 2027

Veer Surendra Sai University of Technology (VSSUT), Burla, Odisha

Relevant Coursework: Data Structures & Algorithms, Database Systems, Artificial Intelligence & Machine Learning, Operating Systems

Fundamentals of Accelerated Computing with CUDA Python — Jul 2025

NVIDIA Deep Learning Institute

Covered: GPU memory hierarchies, parallel thread execution, CUDA kernel design and accelerated array computation with Numba

Leadership & Activities

Cloud & DevOps Lead, Enigma Web & Coding Club, VSSUT — 2024 – Present

- Implemented and maintained CI/CD pipelines for veerpreps.com, a notes platform serving 3,000+ students during exam periods, using GitHub Actions and Docker; mentored 17 junior developers on containerization, AWS deployment and infrastructure best practices

Student Secretary, Department of Computer Science & Engineering, VSSUT — 2025 – Present

- Led planning and execution of departmental initiatives for 1,000+ students and faculty, covering academic coordination, technical events, industry engagement and student welfare across cross-functional teams